Scott Grew

Education

University of Southern California, Viterbi School of Engineering 2026 – 2028
Master of Science in Computer Science

- Fall 2026 coursework: Analysis of Algorithms, Principles of Programming Design
- Spring 2027 coursework: Robotics, Operating Systems, Machine Learning

Orange Coast College Sept 2025 – Aug 2026
Post-Baccalaureate Computer Science and Mathematics Coursework

California State University, Fullerton 2018 – 2022
Bachelor of Arts in Philosophy

Projects

Throughline: Lidar Multi-Object Tracker | *C++, Eigen, Docker*

- Built a live tracker on the Waymo Perception Dataset that follows vehicles and pedestrians through a real detector's boxes, keeping each object's identity through missed scans and forecasting its path 3 seconds ahead
- Scored 0.30 to 0.72 for vehicles and 0.22 to 0.57 for pedestrians on Waymo's official tracking accuracy score (1.0 is perfect) across three driving clips
- Predicted each object's motion with an extended Kalman filter and matched new boxes to those predictions with the Hungarian algorithm, at 2 to 4 milliseconds per frame (under 8 for 99% of frames), well within the 100 milliseconds between scans

MotionNet: Self-Driving Motion Prediction | *Python, PyTorch, Docker*

- Built a 12M-parameter transformer encoder–decoder on 122K Waymo Motion Dataset scenes that uses one second of agent history, nearby traffic, lane geometry, and traffic signals to predict six ranked 8-second trajectories
- Scored all 44.9K held-out test scenes on Waymo's evaluation server, with average best-trajectory errors at 8 s of 1.43 m for vehicles, 0.65 m for pedestrians, and 1.28 m for cyclists, within 0.02 m of the validation split
- Clustered training endpoints into 54 object-specific motion modes, ranked them by confidence, and suppressed near-duplicate predictions

Showdown: Poker Strategy Solver | *C++*

- Built a strategy solver for Rhode Island Hold'em, a simplified two-player poker game used as an AI benchmark
- Computed exactly how much a perfect opponent could win against it: 0.094 chips per hand, 88% lower than the published solution's 0.805 under the same evaluation
- Trained it through self-play with counterfactual regret minimization, on 16 threads over lock-free shared tables, 10× faster than on one thread

Technical Skills

Languages: Bash, C++, Java, Python
Libraries & Frameworks: Eigen, Next.js, NumPy, pandas, PyArrow, pytest, PyTorch
Tools & Platforms: CMake, Docker, Foxglove, Git, Hugging Face, Kaggle, Linux, MongoDB, Protocol Buffers

Leadership & Awards

Micromouse Club, Orange Coast College Feb 2026 – Aug 2026
Founding Board Member

- Ran club lab sessions on maze-solving algorithms and embedded sensors: flood fill, A*, Dijkstra, D* Lite
- Prototyped time-of-flight distance sensing (VL53L1X) on Arduino

HackOC Spookathon Oct 2025
1st Place, Team of Four

- Built a calculus tutor that grades handwritten work from a photo by converting the handwriting to LaTeX and having a language model check each derivation step, flagging the first that does not follow
- Shipped the C++ backend and Next.js frontend in 36 hours as a team of four